TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

TECHNICAL REPORT

Field	Details
Institute / University	PCCOE
Programme / Batch	B.Tech CS/IT/CS-AIML, 2027 batch
Student Name	Om Deshmukh
PRN	123B1C004
Case Study ID (CS1–CS5)	CS1 — Automotive HLD Document Analysis Assistant
Project Title	AutoDoc AI — Automotive Architecture Assistant

1. Problem Statement and Intended Users

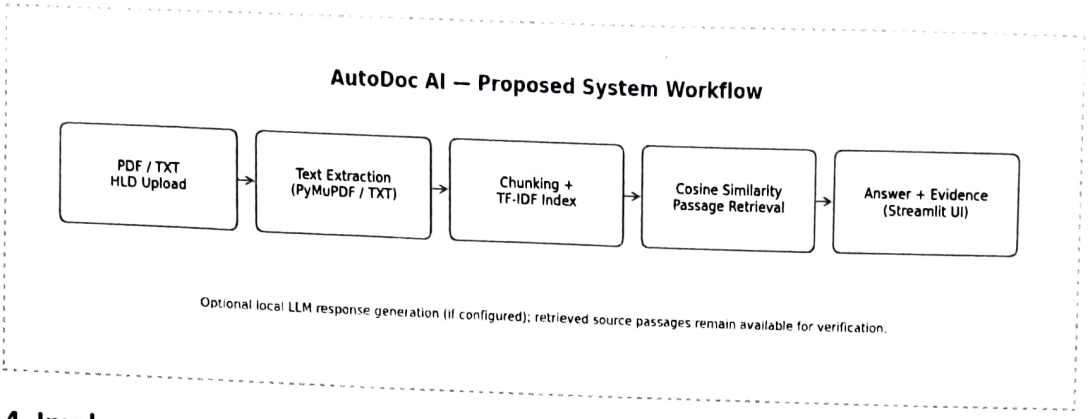
AutoDoc AI addresses the effort required to locate relevant information in automotive High-Level Design (HLD) documents. Intended users include students, engineers, and technical reviewers who need a document-grounded starting point for exploring architecture details. The prototype accepts PDF/TXT files, extracts text, retrieves relevant passages, and displays source/page evidence. Responses use an extractive fallback by default, with optional local Ollama generation. The system is educational and is not a certified AUTOSAR or vehicle-safety tool.

2. Input Data / Knowledge Base

Item	Description
Sources	Input_Data/synthetic_autosar_hld.txt (synthetic illustrative HLD); user may upload an approved PDF or TXT. Sample is not OEM or normative AUTOSAR content.
Size and format	Small UTF-8 text sample; application accepts PDF and TXT. Actual size depends on the uploaded document.
Preprocessing / EDA done	PDF text is extracted page by page using PyMuPDF (fitz); TXT is decoded as UTF-8. Empty PDF pages are skipped. Extracted content can be previewed.
Chunking strategy	Page-aware chunks of 180 words with 35-word overlap.

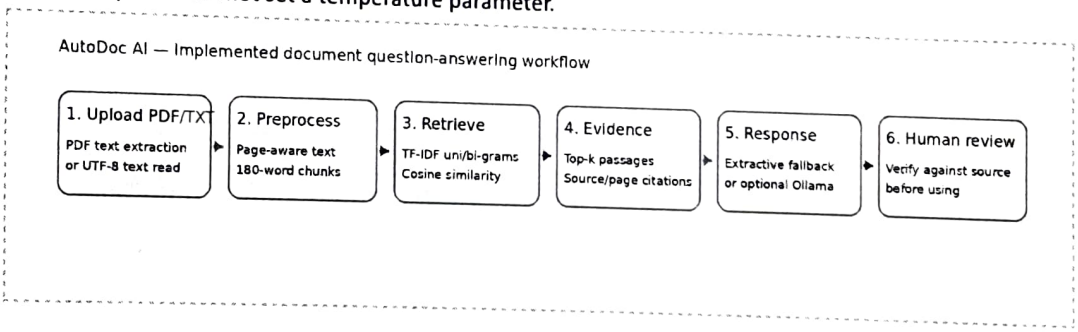
3. Solution Architecture

Component	Choice (name, version, hosting)
Pre-trained model	Optional Ollama local LLM; default model name is llama3.2:3b, disabled by default. Record the actual model/version only if enabled and successfully tested.
Embedding model	None. scikit-learn TF-IDF sparse lexical vectors are used, with unigram and bigram features.
Vector store (FAISS / ChromaDB)	None. Chunks are stored in Streamlit session memory; no persistent vector database.
RAG pipeline / AI-agent design	Ingest PDF/TXT → extract page text → split into overlapping chunks → TF-IDF vectorization → cosine-similarity ranking → display top-k passages and source/page labels → extractive fallback or optional Ollama response.
Service layer and UI	Python and Streamlit; PyMuPDF for PDF extraction; scikit-learn for TF-IDF and cosine similarity.
Tools / deterministic rules	UI default top-k=4, adjustable 1–6; evaluation script uses top-3. Empty/zero-score retrieval is flagged. Optional Ollama endpoint defaults to http://localhost:11434.



4. Implementation and Configuration

The workflow begins when the user uploads a PDF or TXT document. PDF pages are extracted with PyMuPDF; TXT content is decoded as UTF-8. Each page is split into 180-word chunks with a 35-word overlap. Retrieval uses scikit-learn TfidfVectorizer with English stop words, ngram_range=(1, 2), and sublinear_tf=True; cosine similarity ranks chunks. The UI defaults to top-k=4 and allows 1–6 evidence passages. If the user enables Ollama and the local service is available, retrieved context and the question are sent to the configured model; otherwise the app returns an extractive evidence summary. The current Ollama request does not set a temperature parameter.



5. Evaluation Steps and Results

#	Test question / scenario	Expected answer / behaviour	Result (Pass/Fail, notes)
1	Question about a named component/interface	Expected terms occur in relevant top-3 passages.	Aggregate scripted result: all 8 configured questions passed; see generated CSV for per-question details.
2	Question about diagnostic functionality	Retrieve passages containing diagnostic expected terms.	Aggregate scripted result: all 8 configured questions passed; see generated CSV for per-question details.
3	Question about communication behavior	Retrieve the expected bus/interface terminology.	Aggregate scripted result: all 8 configured questions passed; see generated CSV for per-question details.
4	Question with specified expected evidence terms	At least 50% of expected terms occur in top-3 retrieved text.	Pass under the script's lexical criterion for the 8-question set.
5	Question whose answer is absent from the document	System should signal insufficient evidence; manual negative-query test required.	Not included in the numeric score; run and document a manual test before submission.

Metric	Score / observation
Retrieval smoke-test pass rate	8/8 questions passed (100%) on the included synthetic HLD. Pass means at least 50% of expected evidence terms appear in top-3 retrieved text.
Retrieval hit rate / citation accuracy	Not separately measured. Evidence is displayed with filename/page labels; citation correctness needs manual review.
Latency / answer faithfulness	Not measured by the script. The test does not establish answer factuality, robust semantic relevance, safety, or AUTOSAR compliance.

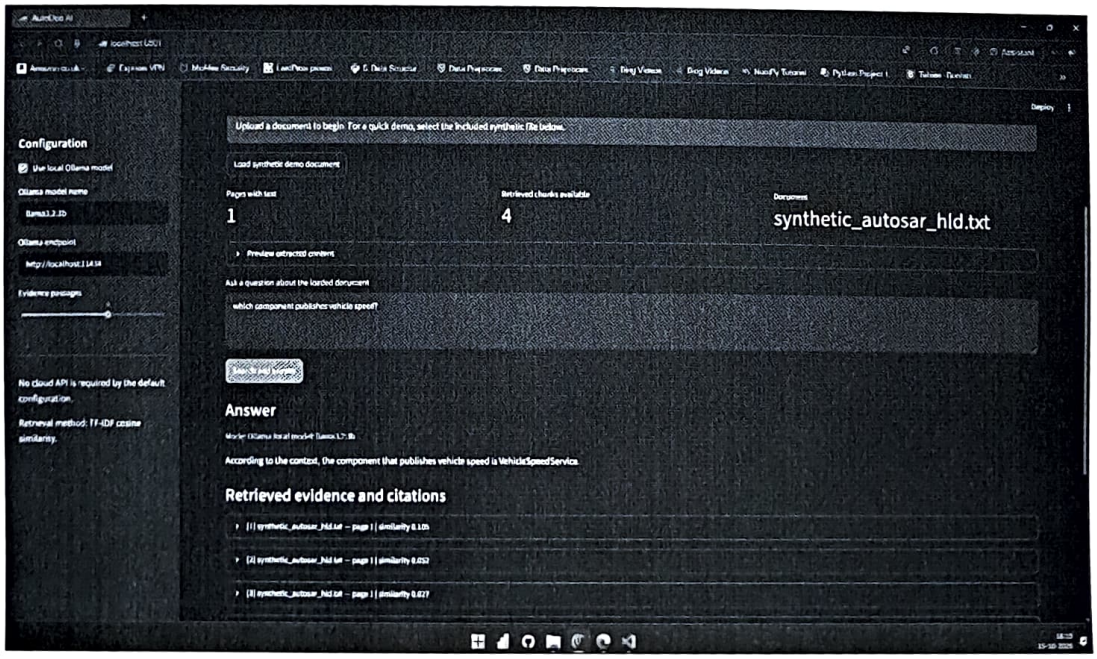
6. Responsible AI Measures

Area	Measure adopted in this project
Grounding and citations	Retrieved evidence is displayed with source filename and page number. Optional Ollama prompt instructs the model to use only supplied context and acknowledge insufficient evidence.
Privacy	Default mode does not call a cloud LLM API. Use only approved or synthetic documents; local processing depends on deployment and configuration.
Security	Prototype has no authentication or role-based access control. Do not expose publicly or process confidential documents without approval.
Human oversight / review	Users must verify responses against source passages and seek qualified review before engineering use. Not production-grade or safety validated.

7. Innovation Highlights

- A reproducible lexical evaluation script reports its metric definition and limitations.
- Extractive fallback is used when an optional local model is disabled or unavailable.
- Page-aware evidence and source labels support traceability.
- Lightweight retrieval works without a cloud LLM API in the default mode.
-

8. Screenshots, Logs and Evidence



9. Reflection and Conclusions

The prototype demonstrates a baseline workflow for document ingestion, chunk-based retrieval, and evidence display. The supplied evaluation script was run against the included synthetic HLD: 8 of 8 configured questions passed, yielding a 100% lexical evidence-term pass rate. This result is limited to a small synthetic dataset and a rule requiring at least half of the expected terms to appear in the top three retrieved passages; it does not measure generated-answer correctness or safety. The system uses TF-IDF rather than semantic embeddings, has no persistent vector store, and does not implement OCR for scanned PDFs. Future work should include a larger reviewed test set, semantic retrieval, explicit negative-query testing, latency tracking, and testing with approved representative documents. Human review remains essential.

10. AI-Tool Usage Declaration

AI tool / model	Purpose	Artifact affected	Verified By Student (Y/N)
ChatGPT / GPT-6 (AI assistant)	Planning, drafting, and coding assistance; verify and disclose as required by institution.	Synopsis, documentation, report text, and code where applicable.	[Y/N]
Ollama local LLM (optional)	Optional answer generation from retrieved context; claim use only if enabled and successfully tested.	Runtime answers only if actually used; record exact model/version.	[Y/N or N/A]

11. Student Declaration

I confirm that this report represents my own understanding and contribution, that the work follows the approved case-study scope, and that I can explain every significant part of the implementation.

Signature:		**Date:**	7/10/2026
Name:	Om Deshmukh	**PRN:**	123B1C004